



UNIVERSITY
OF WOLLONGONG
AUSTRALIA

INTERMARC GLOBAL

ENGG454 – Professional Experience

Student Name	Yasan Semila Gunathilaka
Student Number	7549246
Position Held	Intern

Organisation Name	Intermarc Global Pty Ltd
Mentor Name	Tharaka Madduma Liyanage
Mentor Position	Manager – Engineering Services
Mentor Contact	0447401198
Employment Period	09/12/24 – 28/02/25 (12 weeks)



03TH March 2025

Word Count: 4021 words

I. EXECUTIVE SUMMARY

This report provides an account of my twelve-week internship at Intermarc Global Pty Ltd, detailing the projects undertaken, skills acquired, and their relevance to my Mechatronics Engineering degree. Over the course of the placement, I was involved in three major projects, each contributing to my technical and professional development. This report explores the tasks performed, lessons learned, and how they align with my academic background.

Beyond project work, I gained exposure to industry standards, workplace procedures, and professional practices. These included safety regulations, sustainability protocols, and workplace policies, which are crucial in any engineering role. The report also reflects on my personal growth, evaluating the challenges faced, problem-solving strategies applied, and key takeaways from this experience.

This internship has provided valuable insights into the mechanical engineering consulting sector, equipping me with practical skills and professional exposure that will enhance my transition from academia to industry.

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1. INTRODUCTION

This report documents my twelve-week internship at Intermerc Global Pty Ltd (IMG), undertaken as part of ENGG454, a compulsory professional experience requirement for the Bachelor of Engineering (Honours) – Mechatronics. The placement took place from the 9th of December 2024 to the 28th of February 2025, during which I worked as an Undergraduate Engineering Intern, mentored by Mr. Tharaka Madduma Liyanage, the Manager of Engineering Services.

Intermerc Global is a consulting firm specializing in building services and commercial real estate, offering expertise in mechanical, electrical, and hydraulic engineering. My role involved assisting with various engineering projects, allowing me to develop both technical and professional skills in a real-world environment.

This report provides an in-depth look at the projects I contributed to, the engineering methodologies I engaged with, and the skills I developed. It also includes a reflection on the professional standards, workplace culture, and industry expectations encountered during my internship.

2. ORGANISATION ACTIVITIES

Intermerc Global Pty Ltd (IMG) is a leading engineering consultancy with over 30 years of industry experience. The firm specializes in mechanical, electrical, and civil engineering services and has extensive expertise in property consulting, particularly in commissioning services such as HVAC, Fire, and Building Management Systems (BMS). In addition, IMG provides services in facilities management, Annual Fire Safety Statements (AFSS) coordination, building operations, technical services, compliance, financial management, fit-outs, procurement, and project management.

At the core of IMG's operations is a commitment to delivering projects on time and within budget. Their service model is built on four key principles:

- Highly experienced and motivated technical staff
- Best practice and continuous improvement
- Performance measurement and reporting
- Benchmarking for industry standards

Some of the key projects undertaken by IMG include:

1 Rivett Road, North Ryde (3M Building)

- Lightning upgrade
- BMS (Building Management System) upgrade
- Installation of Optus & Telstra mobile antennas
- Electrical and HVAC system condition assessment

131 York Street

- Lift upgrade project (03 passenger lifts)

2 Wentworth Street – Parramatta

- Upgrade of EWIS, fire indication panel, fire alarm and detection system

377 Sussex Street

- Mechanical services review, improvements and tendering
- Fire damper rectification work
- Façade inspection and repair management
- Preliminary investigation for a solar energy project

Research & Development Initiatives

Beyond consultancy services, IMG places strong emphasis on engineering research and development. The company holds a patent for a revolutionary control system for electric machines, specifically an AC machine controlled via amplitude modulation. This patented technology is currently under further development for commercial manufacturing.

Additionally, IMG is actively engaged in the design and development of an advanced electric motor tailored for the rapidly expanding electric vehicle (EV) and electric aviation industries. The objective is to achieve higher power density and improved efficiency compared to existing solutions, while eliminating the need for permanent magnets, contributing to both cost reduction and sustainability.

3. PROFESSIONAL EXPERIENCE ACTIVITIES

During my tenure at Intermerc Global, I was involved with 3 major projects. A summary of the work that was undertaken, the knowledge and experience gathered, and the relation of the tasks to my degree is provided below.

3.1. Project 1 - Compliance Audit on Kitchen Exhaust System



Figure 1 - Kitchen Exhaust System, 45 Clarence Street

Overview

The first project focused on conducting a compliance audit for the kitchen exhaust system at 45 Clarence Street. Given my limited prior experience in the industry, the project was structured into two key stages:

Stage 1: Understanding Standards and Regulations

To ensure a thorough audit, I first familiarized myself with the Australian Standards and regulations governing mechanical ventilation systems. This involved in-depth research and analysis of the following key documents:

1. AS 1668.1:2015 – The use of ventilation and air conditioning in buildings, focusing on fire and smoke control.
2. AS 1668.2:2012 – Mechanical ventilation requirements in buildings.
3. National Construction Code (NCC) – Regulatory framework governing building construction and safety.

Stage 2: Compliance Assessment and Calculations

Once I had a solid understanding of the standards, I conducted a comparative analysis to determine whether the existing kitchen exhaust system complied with the relevant requirements. This process involved:

- Reviewing mechanical drawings provided by the client.
- Drafting a compliance report outlining standards that were met and highlighting non-compliance areas.
- Performing technical calculations to assess whether the exhaust fan capacity was sufficient for the room volume and required airflow rates.

This project not only improved my understanding of ventilation system design and compliance but also enhanced my ability to interpret technical standards and perform engineering assessments, which are valuable skills in the field of Mechatronics Engineering.

What did I learn?

This project provided invaluable insight into the complexity and rigor involved in adhering to industry standards and regulations. I learned that detailed policies and standards are essential to ensure safety and efficiency, particularly in systems related to fire safety and ventilation. These standards are strictly designed to account for a wide range of scenarios and applications. A prime example is the requirement outlined in AS 1668.2:2012, Section 3 (Mechanical Ventilation – Exhaust Systems), which states:

“When not enclosed in a dedicated fire-resistant shaft or enclosure, ducts shall be made of galvanized steel greater than 1.2 mm or stainless steel greater than 0.9 mm.”

The standard documents also provided formulas to estimate the flow rates for different hood systems. The calculations for the two hood types used in this system were as follows:

- Flow rate (Q) for Hood Type 3 (side wall):

$$Q = 375 \times P \times H = 375 \times (2W + L)$$

$$Q = 375(2(0.97) + (6.107)) = 3,017.625 \text{ L/s}$$

- Flow rate (Q) for Hood Type 4 (island):

$$Q = 190 \times P \times H = 190 \times (2W + 2L)$$

$$Q = 190 \times (2(1.01) + 2(1.01)) = 767.6 \text{ L/s}$$

These calculations helped me assess whether the exhaust fan was adequate for the system's requirements. This experience enhanced my understanding of engineering calculations and system design evaluations.

Correlation to my degree

The calculations performed during this project closely related to concepts I had studied in MECH252 (Thermodynamics, Experimental Methods, and Analysis). The experience of using

the data booklet to extract relevant formulas, such as those for flow rate, allowed me to apply theoretical knowledge to real-world problems.

Additionally, this project emphasized the importance of meeting all standards and regulations—a fundamental aspect of engineering practice. It reinforced how crucial it is to incorporate factors of safety in engineering design and operations, as ignorance of standards can lead to serious consequences. This principle aligns with the professional standards set by Engineers Australia.

While I had not yet taken courses specifically targeting standard practices, I believe subjects like ENGG105 (Engineering Design for Sustainability), ECTE351 (Engineering Design and Management 3), and ENGG461 (Managing Engineering Projects) have deepened my understanding of these regulations and their practical applications. Gaining the ability to interpret and apply these standards is key to becoming an industry-ready engineer.

3.2. Project 2 – 3M Building, 1 Rivett Road



Figure 2 - 3M Building

What was undertaken?

The goal of this project was to introduce me to Building Management Systems (BMS), also known as Building Automation Systems (BAS). These systems centralize control and monitoring of various mechanical, electrical, and other operational systems within a commercial building. The primary purpose of BMS is to enhance efficiency, reduce energy consumption, and optimize operational costs while simplifying the overall management of the building. In the case of the 3M Building, Alerton Systems were implemented to monitor and control the building's infrastructure. As part of the project, I was involved in assisting the IMG team to ensure that the BMS operated smoothly and efficiently.

BMS typically integrates several systems within the building, including:

- HVAC (Heating, Ventilation, and Air Conditioning)
- Lighting
- Fire Safety
- Access Control

These systems work together to create a more energy-efficient and safe environment for the building's occupants.

What did I learn?

Through this project, I was introduced to the major components that form the backbone of a BMS, which include:

- BMS Server (Head End): The main control interface that gathers data and manages all connected systems.
- Main Plant Controller: Manages large-scale building systems like HVAC.
- Terminal Unit Controllers (TUC): Controls specific units like individual air handling units or lighting systems.

- High-Level Interface (HLI): A user-friendly interface for operators to interact with and monitor the system.

My task was to examine technical documentation and study how various sensors and devices perform their roles. For example, temperature sensors detect heat levels, and the data collected is transmitted to the BMS server, providing real-time feedback on building performance.

I was also exposed to the software and communication protocols used within BMS. Key protocols included:

- BACnet: The most widely used protocol in building automation, ideal for large networks due to its auto-discovery capabilities.
- Modbus: A reliable protocol often used for metering and integration with Variable Frequency Drives (VFDs).
- M-Bus: Specializes in water and gas metering, including wireless options.
- DALI: Best suited for lighting systems control.
- KNX: A versatile protocol for lighting and small HVAC systems, also enabling gateway integration into BMS.



Figure 3 - Communication Protocols

A key feature I encountered was Direct Digital Control (DDC), a system used in the 3M building. DDC employs microprocessors to control various building systems in real-time. This system provides higher precision and flexibility compared to analogue control systems. The digital nature of DDC allows for the programming of custom control strategies, leading to more tailored solutions. Additionally, DDC enables predictive maintenance and energy management by collecting and storing data for trend analysis and diagnostics, thus enhancing the operational efficiency of the building.

Correlation to my degree

The core principles of closed-loop control systems that I encountered in BMS are directly related to concepts covered in my Mechatronics Engineering degree. Specifically, the basic theories of control systems, where a command is issued and feedback is recorded to adjust the

system's performance, are fundamental to both BMS and the broader field of control engineering.

In my ECTE344 (Control Theory) subject, I had previously worked with the Quanser QUBE-Servo 3 motor, using MATLAB and Simulink to manipulate a DC servo motor for advanced control tasks. The concepts of feedback loops, error correction, and real-time control that I applied to the QUBE-Servo project are the same principles that underlie BMS and Direct Digital Control (DDC) systems.

This hands-on exposure to BMS helped me see the real-world application of the control theories I studied in my degree. It also allowed me to understand how the integration of different systems (e.g., HVAC, lighting, fire safety) can be controlled through digital interfaces, contributing to building efficiency and sustainability.

3.3. Project 3 – ACA Building, 66 King Street



Figure 4 - ACA Building

What was undertaken?

This project focused on expanding my understanding of larger mechanical systems used in commercial buildings, specifically at the ACA Building. While many of the systems used in the ACA building are like those used in smaller buildings, this project exposed me to more complex and large-scale systems, including additional exhaust systems, boilers, and chillers.

The ACA building spans 15 levels, and the major systems were located on Level 14. Some of the key mechanical components included:

- 2 Air-cooled Carrier chillers
- 2 Chilled water pumps
- 2 Hot water pumps
- Ventilation fans
- Integration with tenancy occupancy relays

What did I learn?

To begin the project, I was granted access to the Building Management System (BMS) folders, which included detailed mechanical and electrical technical drawings. My task was to go through these documents to identify the systems incorporated into the ACA building and understand how they integrated with one another.

I quickly realized that the drawings were much more detailed and complex than I initially expected. I needed guidance from my mentor to help me interpret the layout and systems. Through this, I was able to gradually develop an understanding of how each system worked and interacted across the building.

The mechanical systems involved in this building were larger and more intricate than those I had dealt with previously, making it essential to understand the functionality of chillers and

boilers. The chiller system used at ACA is air-cooled, but I also learned that water-cooled chillers are often used in large-scale buildings. To deepen my understanding, I had to conduct research into how these systems work and compare them with other components like Air Handling Units (AHU) or Rooftop Units (RTU).

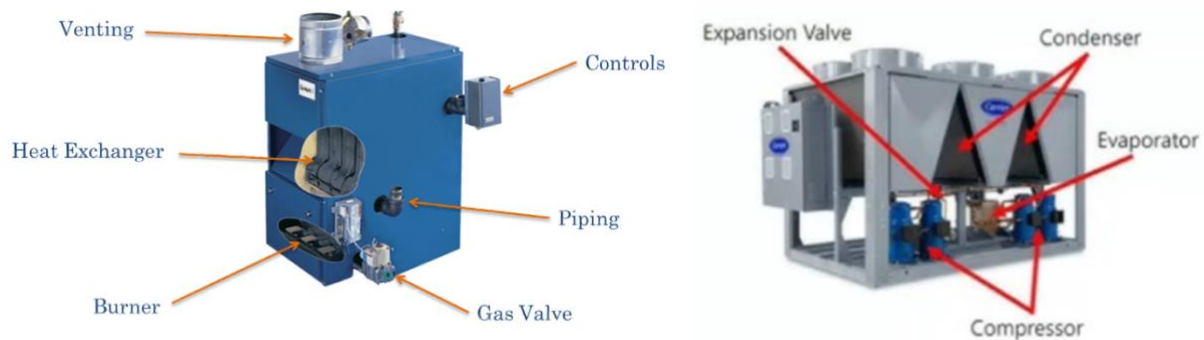


Figure 5 – Diagrams of a boiler (left) and chiller (right)

Additionally, during one of my site visits, I was shown a faulty chiller on Level 10. Observing the troubleshooting process and listening to my mentor's explanations further enriched my understanding of the system's operations. It allowed me to visualize the concepts I had learned through the drawings and networked documents in real time.

A secondary task I worked on was understanding the sprinkler system, particularly its activation mechanism. I explored how dry-pipe and wet-pipe systems function. It was fascinating to learn about the glass bulb/solder plug mechanism that triggers the sprinklers when a certain temperature is reached. This system is designed so that only the sprinklers nearest the fire activate, preventing unnecessary water flow in unaffected areas.

Correlation to my degree

While I had previously learned about pumps in ENGG252 (Engineering Fluid Mechanics) and MECH252 (Thermodynamics, Experimental Methods and Analysis), the operation of chillers and boilers was new to me. To bridge this knowledge gap, I conducted independent research to understand how these systems function.

Although chillers and boilers operate on different principles, I found that many of the analytical methods I had learned in my coursework could still be applied. For example, both chillers and boilers can be analysed using thermodynamic cycles, like the way I would analyze pumps in fluid systems. This connection reinforced the idea that core engineering principles, like thermodynamics and fluid dynamics, are transferable across various systems and real-world applications.

Additionally, this project provided valuable insight into how large-scale mechanical systems are designed, integrated, and maintained within a commercial setting. It underscored the

importance of detailed technical documentation and the ability to interpret complex systems - a skill that will prove critical in my future engineering career.

3.4. Other Activities

What was undertaken?

In addition to my primary projects, I was tasked with smaller theory-based activities to broaden my understanding of various systems used in the industry. At Intermerc Global, engineers are expected to be knowledgeable about a wide range of systems, such as elevators, fire dampers, active and passive fire protection systems, and more. To gain a better grasp of these systems, I conducted thorough research, and here is a summary of what I learned.

Traction Elevators

Traction elevators use a counterweight attached to the elevator cab to reduce the load on the motor. This counterweight is typically about half the weight of the cab when loaded. Traction elevators are equipped with multiple safety mechanisms to ensure the safe operation of the elevator system. These include:

- Machine brake: A brake within the motor that stops the motor from turning in emergencies.
- Tension cables: Multiple cables (4-8) capable of handling the entire weight of the cab.
- Emergency cab brake: A brake that locks the cab to the rails when there is a rapid descent, triggered by spinning weights that activate if the speed exceeds a set threshold.
- Door mechanism: Each floor has a mechanical interlock that ensures the floor door only opens when the elevator door is aligned properly.

Fire Dampers

Fire dampers are crucial components in maintaining safety in a building during a fire. These devices automatically close when heat is detected, preventing the spread of flames through air ducts in fire barriers. Fire dampers typically include a fusible link that melts when exposed to high temperatures, causing the damper blades to close and restrict the flow of fire.

Smoke Exhaust Systems

Smoke exhaust systems are designed to remove smoke from a building during a fire, creating pressurized zones to isolate the fire and prevent its spread. These zones are kept under negative pressure, while floors above and below the fire zone may be kept under positive pressure. In the case of fire exits, stairway pressurisation fans are used to ensure smoke is kept out of stairwells while still allowing people to enter. Additionally, both supply and exhaust fans used in smoke exhaust systems are required to be fire-rated, meaning they can function for a minimum of two hours under high heat and pressure conditions.

Active vs. Passive Fire Protection Systems

While active and passive fire protection systems both aim to keep a building safe from fire, they function in different ways:

- Active systems are designed to detect, suppress, and help people escape from a fire. Examples include sprinklers, smoke detectors, and fire extinguishers.
- Passive systems focus on preventing the spread of fire. These include fire doors, fire curtains, and fire & smoke dampers.

Heat Pumps & Air Conditioning Units

In my studies for MECH252 (Thermodynamics, Experimental Methods and Analysis), I had encountered heat pumps and air conditioning units, but I never fully understood how they could achieve an efficiency greater than 100%. Through additional research, I gained clarity on how these systems work. Unlike boilers, which generate heat, heat pumps and air conditioners simply move heat from one place to another using a refrigerant and fans. This process doesn't create heat, but rather transfers existing heat, which is why they can be more efficient than other heating methods. This process adheres to the laws of thermodynamics, particularly the principles of energy conservation and the coefficient of performance (COP).

4. STUDENT EVALUATION

Working at Intermerc Global has been a significant learning experience, providing a valuable insight into the workings of a professional engineering company. While I have prior experience working in customer service, this internship exposed me to the more technical and collaborative aspects of a professional environment, which differ greatly from my past roles.

One of the most rewarding aspects of my time at Intermerc Global was the opportunity to work alongside a team of experienced engineers. Beyond the specific projects and activities assigned to me, I gained knowledge simply by being involved with the team and listening to their discussions and meetings. Their expertise was not only evident in their technical work but also in their approach to problem-solving and project management.

I spent a considerable amount of time working closely with my mentor, Deshan Liyanage (Manager of Engineering Services) and key staff members like Chanuka Wijayasiri (Manager of Support Services) and Dhammika Gunaratne (Managing Director). The team was always approachable and eager to assist when I encountered challenges or had questions. This supportive environment helped me feel comfortable seeking guidance whenever necessary. The office culture was both professional and friendly, making it easy to integrate into the team and learn from others, even during breaks and after hours.

Throughout my internship, I was primarily focused on three key projects: the Kitchen Exhaust Compliance Audit, 3M Building, and ACA Building. These projects provided me with exposure to a broad range of systems, including Mechanical, Electrical, HVAC, Fire and Building Management Systems (BMS). Although the latter two projects are ongoing and handled by other team members at IMG, I contributed by drafting reports (such as the Compliance Audit) that served as helpful references for the team and summarized key aspects of each project.

Initially, I found it challenging to familiarize myself with the technical jargon and engineering drawings associated with these projects. Due to my limited prior experience in these areas, it took me time to understand the systems involved, such as boilers, chillers, elevators, exhaust systems, fire dampers, and sprinklers. Personal study was necessary to gain the required background knowledge and fully comprehend the operations of these systems.

Overall, my internship experience has given me a clearer picture of what to expect when working in an engineering company. It has built my confidence and improved my ability to ask questions when I am unsure about a task or how to proceed with my work. Additionally, it has emphasized the importance of teamwork and the value of collaborating with colleagues who can provide feedback to ensure the highest quality work is delivered to clients.

5. PROCEDURES WITHIN ORGANISATION

At Intermarc Global, safety and security are top priorities, and strict adherence to standard engineering practices ensures the well-being of all personnel. This commitment to safety is evident in the organisation's approach to both workplace health and safety, as well as physical and data security measures.

Safety

During the induction process, I was introduced to the company's comprehensive safety procedures. The workspace is designed with clear distinctions between rooms for specific purposes, such as the office, workshop, meeting room, kitchen, and toilets. Each room is separated by barriers or walls to ensure safety and prevent interference of activities.

Mechanical and electrical equipment is securely stored in the workshop, with access restricted to trained and qualified staff. Each piece of equipment is carefully placed in its designated area, accompanied by appropriate warning signs and hazard labels (e.g., for high voltage) to alert staff to potential dangers.

All staff members and long-term visitors are briefed on fire safety and evacuation procedures. This includes clearly marked fire exits and assembly areas. The building is equipped with signage to guide personnel in and out of the office, as well as road signs and lane markings to direct both pedestrians and vehicles safely.

For site visits, staff are encouraged to travel in teams to enhance safety. This approach ensures support during navigation and fosters a safer work environment for all personnel. As an intern, I was always accompanied by my mentor or another experienced staff member. The whereabouts of all personnel during site visits are reported back to the office to maintain oversight.

Security

Access to the office is controlled via an access card system, with a single entrance for staff and visitors. The building is equipped with CCTV to monitor and record the movement of personnel entering and exiting the premises. On the sites I visited, access was strictly limited to authorised individuals, often requiring special keys stored in safes to access critical areas such as the building management control systems. Additionally, security protocols like disabling alarm systems were in place to control entry.

The control systems employed also feature comprehensive history logs, tracking the time, reason, and individual responsible for accessing restricted areas. This allows the company to maintain accountability and security. In terms of data protection, the organisation employs multiple layers of security to safeguard sensitive client information. Access to these files is restricted and requires separate login credentials, with documents protected by two-factor authentication to prevent unauthorised access.

Sustainability

Intermarc Global is committed to implementing strategies that support sustainability and work towards a net-zero carbon footprint for future profitability. A key initiative in this effort is the adoption of IoT-driven monitoring and optimisation of building operations, reducing the need for engineers to travel to sites in person. By integrating remote access and control systems for heating, lighting, and security, the company aims to enhance operational efficiency and costs while minimising its environmental impact.

6. APPENDIX

A. Unpaid Work Experience Application Form

09/12/2024, 10:55

Unpaid Work Experience Application v1



UNIVERSITY
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AUSTRALIA

Engineering and Information Sciences Unpaid Work Experience Application

Student Details

Name	Yasan Gunathilaka
ID Number	7549246
Degree Course	Bachelor of Engineering (Honours) (Single Major)
Discipline	Mechatronic Engineering
Year of your degree	3rd year
Email	ysg631@uowmail.edu.au
Session Intending to Graduate	Spring
Year Intending to Graduate	2025
Any Pre-Existing Medical Conditions or Information Relevant to the Placement	No

Period of Employment

Start Date	12/9/2024
End Date	2/28/2025
Full time or part time employment	Please specify in format (x days/wk * y wks = x*y days) eg. (3 days/wk * 20wks = 60 days) 5 days/wk * 12 wks = 60 days
Schedule of employment	eg. 12 weeks or 6 + 6 weeks 12 weeks
Comments for period of employment	During the christmas holidays, the student would be given a project to work on their own time, while consulting the organisation mentor

Organisation Details

Name of Organisation	Intermarc Global Pty Ltd	Website	http://intermarcglobal.com.au
Address 1	30/1 Rivett Rd	State	NSW
Address 2		Postcode	2113
Suburb	North Ryde	Country	Australia

Organisation Student Mentor Details

First Name	Tharaka	Professional Qualification	Masters in Mechanical Engineering, Bachelor of Science in Mechanical & Manufacturing Engineering, Professional Member - Engineers Australia,
Last Name	Madduma Liyanage	Position and Dept. at Organisation	Manager - Engineering Services
Email Address	deshan@intermarcglobal.com.au	Contact Phone	0447401198

Student's Research into Organisation

The following criteria will be used to check if the organisation and placement you are submitting can satisfy the requirements

Organisation Profile

Product/activities, number of engineers employed, history of the company.

Note: a medium/large company is preferred

Experienced engineering consultants (Mechanical, Electrical & Civil) in Australia's building services and commercial real estate industries, with extensive expertise in engineering and asset management. Over 30 years of combined experience and knowledge of the industry. Projects include: Life cycle and condition assessment projects, Annual Fire Safety Statement (AFSS) Coordination, Power Engineering & Building Management. Partnering with NCINGA enhances our capabilities with IoT and data analytics. Our collaboration delivers innovative solutions that boost the performance and efficiency of our clients' properties. Our services are centered on: Modernising electrical, mechanical, security and smart systems. IoT-driven monitoring and optimisation of building operations. Using automation and AI for efficient building management and data storage... Research and Development: IMG holds a patents for one fo the most advanced control systems for electric machine (an AC machine controlled via Amplitude Modulation) in IP Australia & US Patent. Our research and development team is spearheading the design and engineering of an electric motor tailored specifically for the evolving needs of electric vehicles (EV) and electric flying objects.

Work Experience Expected

The work experience needs to be related to an **Engineering** problem. It is expected that the student will be involved in significant aspects of the project(s) assigned by the industry.

Simple site visit is NOT accepted. Work experience needs to ensure the student is trained to be a future engineer, not a skilled worker.

Hands on experience related to engineering projects in HVAC, Fire, Mechanical, Electrical, and Building Management System will be provided. Training related to integration of engineering within facilities management in commercial buildings will be provided. An insight into the research and developement of the electric motor (patented) engineered without the use of permanent magnets, including it's high-precision control system.

Additional projects assigned to student to be completed by the end of internship: 1) Compliance audit on kitchen exhaust system of 45 Clarence street, Sydney 2) Condition assesment for Building Managment System (BMS) of 66 King street, Sydney

Student Declarations

Yes	As the student, I declare I have had initial discussions with this organisation. I declare I have had previous discussions with the contact person at this organisation regarding a professional experience placement, and this contact person will be aware of this communication via an email link to complete the form.
Yes	As the student, I declare there is no conflict of interest for this work placement For example, the student can NOT be supervised by his/her relatives. Note: Self-employed company does not account for work experience.
Yes	As the student, I declare this work placement is UNPAID • I am not employed by the Placement Organisation and will not be receiving payment in respect of the Placement. • I agree that I will only perform activities that fall within the scope of the Brief Description of Placement Activities described above. If I am asked to do other activities I will first notify the University to obtain approval to do so. • I will attend my Placement only during the Placement Dates above. If these dates change I will notify the University. I confirm that I have read the latest Product Disclosure Statement for Personal Accident Insurance located here and I have disclosed all information relevant to my ability to participate in my Placement including any pre-existing medical conditions. I indemnify and hold harmless the University from any loss, expense or damage suffered or incurred by it as a consequence of my failure to make a true and full disclosure of any relevant information, including information relating to any pre-existing medical conditions I have.

Additional Student Comments

Add any general comments regarding this application

Actioned by Yasan Gunathilaka (7549246) on 12/9/2024 10:55:23 AM

Has not been actioned

UOW Insurance has been approved for the above student and placement. ☒ Yes ☐ No
Certificate of Insurance

B. Organisation Mentor – Certificate of Service



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Faculty of Engineering and Information
Sciences

Organisation Mentor's Report - Certificate of Service

Student Details

Student's Full Name	Yasan Gunathilaka	Degree Course	Bachelor of Engineering (Honours) (Single Major)
Student's UOW Email	ysg631@uowmail.edu.au	Discipline	Mechatronic Engineering
UOW Student Number	7549246	Year in degree	3rd year

Organisation Details



Name of Organisation	Intermarc Global Pty Ltd	Mentor's First Name	Tharaka
Address 1	30/1 Rivett Rd	Mentor's Last Name	Madduma Liyanage
Address 2		Mentor's Email	deshan@intermarcglobal.com.au
Suburb	North Ryde	Mentor's Phone	0447401198
State	NSW	Mentor's Professional Qualification	Masters in Mechanical Engineering, Bachelor of Science in Mechanical & Manufacturing Engineering, Professional Member - Engineers Australia,
Country	Australia	Mentor's Position in Organisation	Manager - Engineering Services

Completed Final Report

Either Upload document file (max file size 30 MB)

IMG Final Report.pdf

OR URL to access document online

Actioned by Yasan Gunathilaka (7549246) on 28/02/2025 7:38:08 PM



Placement Details

Placement Start Date	9/12/2024
Placement End Date	28/02/2025
Placement Duration Comments	

Organisation Mentor's Feedback

Please tick the appropriate rating for this student and provide optional comments for each attribute

Attribute	Rating	Optional Comment
Technical Ability	☒ Excellent ☐ Average ☐ Poor	Has a sound technical knowledge. Quickly understand the technical work related to industry and apply the knowledge obtained from the university for practical scenarios..
Initiative	☒ Excellent ☐ Average ☐ Poor	
Cooperation	☒ Excellent ☐ Average ☐ Poor	Equipped with good communication skills. Collaborate with co workers effectively.
Standard of Work	☒ Excellent ☐ Average ☐ Poor	Always delivered the to the standards requested by the company.
Conduct	☒ Excellent ☐ Average ☐ Poor	



Organisation Mentor's Comments

General comments *(including if the report is an accurate description of the work undertaken)*

Mainly three projects were assigned to him. They are a compliance audit on a Kitchen Exhaust System, a BMS audit and a mechanical system review of building. The works undertaken by him and the knowledge obtained are given in the report. During the internship, we involved him for works that is related to the degree and for the fields he will be working in the future.

Actioned by Tharaka Madduma Liyanage on 14/03/2025 11:32:56 AM

C. Weekly Diary

28/02/2025, 19:19

ENGG454 - Student Weekly Diary



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Faculty of Engineering and Information
Sciences

Student Weekly Diary

Name	Yasan Gunathilaka	Organisation	Intermarc Global Pty Ltd
Id Number	7549246	Organisation Mentor	Tharaka Madduma Liyanage
Degree Course	Bachelor of Engineering (Honours) (Single Major)	Placement Start Date	9/12/2024
Discipline	Mechatronic Engineering	Placement End Date	28/02/2025

Weeks	Date	List of Activities	Supporting Documents or Images
Week 1	9/12/2024	The first week of my internship was all about getting acquainted with the team and familiarizing myself with the company's operations. I met everyone at the office, learned about their roles, and gained an understanding of how each person contributes to the company's success. It was also essential to learn about the background of the projects undertaken by Intermark Global (IMG), understanding how the company had successfully completed similar projects in the past. I was introduced to the safety procedures and regulations to be followed at the office, such as evacuation protocols during emergencies, including fires. My schedule for the internship was organized, and I was briefed on the three major projects I would be working on. This included getting familiar with the software tools I would use throughout my placement and receiving a tour of the office facilities. As part of the official initiation process, I completed all the necessary documents and forms to formally start the internship. I was also introduced to Australian Standards, particularly AS 1668 and the National Construction Code (NCC), and began a study on these standards. I wrote a report summarizing what I learned, which would serve as a useful reference tool for both myself and the company when analyzing relevant projects.	

Week 2	16/12/2024	During the second week, I began working on my first project: a Compliance Audit for the Kitchen Exhaust System at 45 Clarence Street. I spent time at the site conducting observations and documenting my findings, including images that were attached in Site1.pdf. The primary task was to draft a detailed report outlining the requirements for a kitchen hood as per the Australian Standards, and then compare the actual site conditions to these standards. This was to ensure that the installation complied with all relevant regulations. I used AutoCAD 2025 to draw the kitchen floor plan (see kitchen_layout.pdf), which was part of the audit process. My mentor provided guidance on the next steps, specifically in relation to calculating whether the exhaust fan used in the system could withstand the air and pressure it would experience during operation. I used a Pressure vs. Flow rate graph to conduct a further comparison. Progress on this project was halted as we were waiting for mechanical drawings from the client to move forward.	kitchen_layout.pdf, Site1.pdf, Compliance Audit on Kitchen Exhaust System.docx, kitchen_layout.bak
Week 3	23/12/2024	The third week marked the beginning of my introduction to Building Monitoring and Control Systems (BMS), an essential aspect of the company's projects. To better understand BMS, I visited a local project at 1A Rivett Rd, North Ryde (3M Building), where I explored the Alerton system in use. This site visit helped me develop a clearer understanding of how BMS systems function, and this knowledge would be crucial as I prepared for the more complex BMS tasks in my second project. In addition to the site visit, I was asked to conduct a literature review and begin drafting a report on the use of BMS and the various systems involved. My mentor shared videos and presentations to assist with my learning, and I began comparing the systems I encountered at 1A Rivett Rd with other BMS systems I would eventually work with during the internship. During office closure for break, I was given the flexibility to continue working on my project from home.	bms-training-presentation.pdf

Week 4	30/12/2024	The fourth week was a continuation of my work on the BMS report. I spent the time drafting and refining the report while also comparing the systems at 1A Rivett Rd with other active IMG projects. This period also overlapped with the Christmas and New Year's break so the office was closed. I was still however able to proceed working on my project at home, which allowed me to focus more on the report without the distraction of other ongoing tasks. I took the opportunity to deepen my understanding of the building management systems, integrating both theoretical and practical knowledge from the site visits.	CM Building Monitoring and Control System.docx
Week 5	6/01/2025	In Week 5, I transitioned to analyzing the BMS system used at 66 King Street (ACA Building), focusing on the mechanical systems integrated into the building's BMS. I drafted a report summarizing the HVAC systems and BMS integration, and spent a considerable amount of time learning how to interpret technical mechanical drawings of the 15-level commercial building. This was an essential skill for understanding how each floor's system was connected and controlled. I also took the time to learn about chiller systems, particularly the differences between air-cooled and water-cooled chillers, and how these systems were integrated into the building's operations.	
Week 6	13/01/2025	Week 6 was busy with two major site visits. The first visit was to 66 King Street (ACA Building), where I assessed the condition of a faulty chiller on Level 10. This gave me hands-on experience with HVAC troubleshooting and system analysis. The second site visit was to 69 Nicholson Street in St Leonards, where I participated in a walkthrough of the building and was involved in arming the security alarm system. During these visits, I had the opportunity to clarify any technical questions I had about the structure and systems of the ACA Building, particularly regarding the design of Level 14.	ACA Level 14.png

Week 7	20/01/2025	This week was dedicated to understanding boiler systems. I focused on the two boilers installed on Level 14 of the ACA Building, learning about their mechanical operation and how they were integrated into the building's BMS. A side project was also undertaken, where I researched fire sprinkler systems, particularly the differences between dry pipe and wet pipe systems and how these systems are activated during a fire. Understanding these systems was crucial for integrating them into the broader building management strategies.	
Week 8	27/01/2025	Week 8 saw me diving deeper into various mechanical systems and their integration into Building Management Systems. IMG engineers frequently encounter a range of devices and systems during their projects, and I was tasked with gaining a comprehensive understanding of how they function. This included: - traction elevators - fire dampers - active and passive fire protection systems - smoke exhaust systems - heat pumps and air conditioning units Each of these systems plays a critical role in the overall safety and efficiency of a building, and understanding their operations was key to my work.	66 King Street.docx
Week 9	3/02/2025	In Week 9, I turned my attention to lift systems and their control logic. I specifically analyzed the old Otis lift system, which used relays for its logic instead of modern computer code. This was an interesting aspect of my learning, as I gained insight into how relays can be used to control elevator systems. I also explored more complex systems, such as duplex lift systems, where multiple lifts are coordinated to operate together. The complexity of relays and how they are integrated into lift operations was an interesting challenge, and I learned how to troubleshoot and optimize such systems.	

Week 10	10/02/ 2025	During Week 10, I began drafting my final report, integrating all the work I had done over the course of the internship. This report included summaries of the Kitchen Exhaust System, 3M Building BMS, and ACA Building projects. I also took the time to document safety procedures, hazard warnings, and other relevant organizational protocols. Photos were taken throughout the internship to capture these procedures, which would later be included in the report section on "Procedures within Organisation."	
Week 11	17/02/ 2025	Week 11 was a follow-up period, where I revisited the Kitchen Exhaust System project. My mentor asked to review the compliance audit report, and I spent time refining and finalizing it. Unfortunately, I was unable to work for a few days due to a throat infection, so I had to take sick leave and missed out on some tasks that week.	
Week 12	24/02/ 2025	In the final week of the internship, I spent time reflecting on the experience and preparing to submit my reports and final tasks. This week focused on tying up loose ends and ensuring that all documentation was completed to the satisfaction of both the company and the university. I caught up with the team to provide feedback on my internship experience, and I returned access to organizational documents as part of the standard handover procedures. The team also briefed me on the final steps to complete my internship, including the communication required with the University of Wollongong (UOW) to fulfill the requirements for ENGG454. Lastly, I provided my own feedback, which was recorded to help improve future internship experiences.	

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